

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Executed preventative maintenance and repair of advanced electromechanical semiconductor equipment at customer sites, ensuring tools returned to optimal performance within an average of **4 hours**.
- Troubleshoot and resolved complex electromechanical issues on semiconductor manufacturing equipment, maintaining a **95% first-time fix rate** for critical components.
- Collaborated with regional engineering teams to escalate and resolve technical challenges, contributing to a **15% reduction** in recurring equipment downtime.
- Developed Python scripts for automating data extraction and analysis from equipment logs, reducing manual reporting time by approximately **10 hours per week**.
- Managed incident response for production systems, performing log analysis and root cause identification using **Splunk** and **Dynatrace**, resolving P1/P2 incidents within SLA targets.
- Participated in cross-functional team meetings, providing technical insights and contributing to the implementation of process improvements for system stability.

SKILLS

Semiconductor Equipment – Installation, Decommissioning, Preventative Maintenance, Repair, Troubleshooting, Electromechanical Systems, Vacuum Control, Gas Control, RF Plasma Systems, Electropneumatic Controls, Microprocessor Controllers, Robotics

Engineering Tools & Technologies – Python, Java, Spring Boot, SQL, REST APIs, AWS (EC2/S3/Lambda/CloudWatch), Docker, Linux, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

Data Analysis & Visualization – Power BI, Tableau, DAX, Power Query, R/RStudio, pandas, NumPy, Statistical Analysis, Data Modelling

Field Service Operations – Customer Training, Technical Documentation, Site Support, Cross-functional Collaboration, Safety Protocols, Cleanroom Standards

Core Engineering Concepts – Mechanical Troubleshooting, Electrical Troubleshooting, Schematics (Electrical/Gas), Problem-Solving, Organization, Communication

PROJECTS

Automated Equipment Diagnostics Tool (HCL Technologies)

- Developed a Python-based diagnostic tool to analyze logs from electromechanical semiconductor equipment, identifying failure patterns that led to a **20% decrease** in recurring issues.
- Integrated the tool with **Splunk** for real-time log ingestion and analysis, enabling faster root cause identification for critical incidents.
- Designed and implemented a user-friendly interface using Flask, allowing field service engineers to run diagnostics and generate reports with minimal training.

Production Line Uptime Monitoring Dashboard (HCL Technologies)

- Engineered a **Power BI** dashboard to visualize real-time equipment status and uptime metrics for a financial services client's operations center, supporting approximately **2,500 transactions** daily.
- Developed complex **DAX** measures and Power Query transformations to aggregate data from multiple sources, providing actionable insights into production bottlenecks.
- Collaborated with business analysts to define key performance indicators and present findings to stakeholders, improving operational awareness and response times by an estimated **18%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)